EXERCISE 11.5

Integrate the following functions with respect to x :

(1) $\dfrac{x^3+4x^2-3x+2}{x^2}$ (2) $\left(\sqrt{x}+\dfrac{1}{\sqrt{x}}\right)^2$ (3) $(2x-5)(36+4x)$

(4) $\cot^2 x+\tan^2 x$ (5) $\dfrac{\cos 2x-\cos 2\alpha}{\cos x-\cos\alpha}$ (6) $\dfrac{\cos 2x}{\sin^2 x\cos^2 x}$

(7) $\dfrac{3+4\cos x}{\sin^2 x}$ (8) $\dfrac{\sin^2 x}{1+\cos x}$ (9) $\dfrac{\sin 4x}{\sin x}$

(10) $\cos 3x\cos 2x$ (11) $\sin^2 5x$ (12) $\dfrac{1+\cos 4x}{\cot x-\tan x}$

(13) $e^{x\log a}e^x$ (14) $(3x+4)\sqrt{3x+7}$ (15) $\dfrac{8^{1+x}+4^{1-x}}{2^x}$

(16) $\dfrac{1}{\sqrt{x+3}-\sqrt{x-4}}$ (17) $\dfrac{x+1}{(x+2)(x+3)}$ (18) $\dfrac{1}{(x-1)(x+2)^2}$

(19) $\dfrac{3x-9}{(x-1)(x+2)(x^2+1)}$ (20) $\dfrac{x^3}{(x-1)(x-2)}$

11.7.3 Method of substitution or change of variable

The method of substitution in integration is similar to finding the derivative of function of function in differentiation. By using a suitable substitution, the variable of integration is changed to new variable of integration which will be integrated in an easy manner.

We know that, if u is a function of x then $\dfrac{du}{dx}=u'$.

Hence we can write $\int f(u)u'dx=\int f(u)du$

Thus, $\int f[g(x)]g'(x)dx=\int f(u)du$, where $u=g(x)$

The success of the above method depends on the selection of suitable substitution either $x=\phi(u)$ or $u=g(x)$.

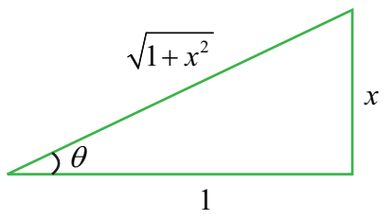
Note 11.2

The substitution for the variable of integration is in trigonometric function, use a rough diagram to find the re -substitution value for it. Suppose the variable of integration x is substituted as $x=\tan\theta$. After integration suppose the solution is $\sec\theta+\operatorname{cosec}\theta$

For example, if $x=\tan\theta$, then from the figure

$\operatorname{cosec}\theta=\left(\dfrac{\sqrt{1+x^2}}{x}\right)$,

$\sec\theta=\left(\dfrac{\sqrt{1+x^2}}{1}\right)$



Then $\sec\theta+\operatorname{cosec}\theta=\sqrt{1+x^2}+\left(\dfrac{\sqrt{1+x^2}}{x}\right)$.

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